

ASSIGNMENT ON CSE-330

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Section: 27

Assignment No: 1

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Answer to the Question No. 1

(a)

$$\beta = 2$$

$$m = 4$$

$$\text{Range of } e = \{-2, -1, 0, 1, 2\}$$

We know standard form $\rightarrow (0.d_1 d_2 d_3)_2 \times 2^e$

(i) Without negative support, Decimal

$$\min = (0.1000)_2 \times 2^{-2} \rightarrow 0.125$$

$$\max = (0.1111)_2 \times 2^2 \rightarrow 3.75$$

(ii) With negative support,

$$\min = -(0.1111)_2 \times 2^2 \rightarrow -3.75$$

$$\max = +(0.1111)_2 \times 2^2 \rightarrow 3.75$$

For Normalized form $\rightarrow (0.1 d_1 d_2 d_3) \times 2^e$

(i) Without negative support, Decimal

$$\min = (0.1000)_2 \times 2^{-2} \rightarrow 0.125$$

$$\max = (0.1111)_2 \times 2^2 \rightarrow 3.875$$

(ii) With negative support

$$\min = -(0.1111)_2 \times 2^2 \rightarrow -3.875$$

$$\max = +(0.1111)_2 \times 2^2 \rightarrow +3.875$$

For Denormalized form $\rightarrow (1.d_1 d_2 d_3)_2 \times 2^c$

(i) Without negative support Decimal

$$\min = (1.0000)_2 \times 2^{-2} \rightarrow 0.25$$

$$\max = (1.1111)_2 \times 2^2 \rightarrow 7.75$$

(ii) With negative support

$$\min = -(1.1111)_2 \times 2^2 \rightarrow -7.75$$

$$\max = +(1.1111)_2 \times 2^2 \rightarrow +7.75$$

(b)

How many numbers can this system represent

(i) Standard $\rightarrow (2 \times 2^3 \times 5) \rightarrow (\text{considering } + \& -)$
 $= 80$

(ii) Normalized $\rightarrow (2 \times 2^1 \times 5)$ "

(iii) Denormalized $(2 \times 2^1 \times 5)$ "

$$= 160$$

(f)

We are required to find the decimal numbers without negative numbers for standard form

For, $e = -2$

$$= (0.1000)_2 \times 2^{-2} = 0.125$$

$$= (0.1001)_2 \times 2^{-2} = 0.125$$

$$= (0.1010)_2 \times 2^{-2} = 0.1875$$

$$= (0.1100)_2 \times 2^{-2} = 0.3125$$

$$= (0.1110)_2 \times 2^{-2} = 0.4375$$

$$= (0.1111)_2 \times 2^{-2} = 0.5$$

$$= (0.1011)_2 \times 2^{-2} = 0.25$$

$$= (0.1101)_2 \times 2^{-2} = 0.375$$

For, $e = -1$

$$= (0.1000)_2 \times 2^{-1} = 0.25$$

$$= (0.1001)_2 \times 2^{-1} = 0.375$$

$$= (0.1010)_2 \times 2^{-1} = 0.5$$

$$= (0.1100)_2 \times 2^{-1} = 0.75$$

$$= (0.1110)_2 \times 2^{-1} = 1$$

$$= (0.1111)_2 \times 2^{-1} = 1.125$$

$$= (0.1011)_2 \times 2^{-1} = 0.625$$

$$= (0.1101)_2 \times 2^{-1} = 0.875$$

For, $e = 0$

$$= (0.1000)_2 \times 2^0 = 0.5$$

$$= (0.1001)_2 \times 2^0 = 0.75$$

$$= (0.1010)_2 \times 2^0 = 1$$

$$= (0.1100)_2 \times 2^0 = 1.5$$

$$= (0.1110)_2 \times 2^0 = 2$$

$$= (0.1111)_2 \times 2^0 = 2.25 \rightarrow 2.25$$

$$= (0.1011)_2 \times 2^0 = 1.25 \rightarrow 1.25$$

$$= (0.1101)_2 \times 2^0 = 1.75 \rightarrow 1.75$$

For $e = 1$

$$= (0.1000)_2 \times 2^1 = 1$$

$$= (0.1001)_2 \times 2^1 = 1.25$$

$$= (0.1010)_2 \times 2^1 = 1.5$$

$$= (0.1100)_2 \times 2^1 = 2$$

$$= (0.1110)_2 \times 2^1 = 2.5$$

$$= (0.1111)_2 \times 2^1 = 2.75$$

$$= (0.1011)_2 \times 2^1 = 1.75$$

$$= (0.1101)_2 \times 2^1 = 2.25$$

for, $e = 2$

$$= (0.1000)_2 \times 2^2 = 2$$

$$= (0.1001)_2 \times 2^2 = 2.5$$

$$= (0.1010)_2 \times 2^2 = 3$$

$$= (0.1100)_2 \times 2^2 = 4.0$$

$$= (0.1110)_2 \times 2^2 = 5.0$$

$$= (0.1111)_2 \times 2^2 = 5.25$$

$$= (0.1011)_2 \times 2^2 = 3.5$$

$$= (0.1101)_2 \times 2^2 = 4.5$$

Answer to the Question No. 2

$$x^2 - 10x + 3 = 0$$

(a)

$$x = \frac{10 \pm \sqrt{100 - 12}}{2} = 5 \pm \sqrt{22}$$

$$x_1 = 9.6904$$

$$x_2 = 0.3096$$

(b)

When we calculate 2nd root

$$x_2 = \frac{10 - \sqrt{88}}{2}$$

So, we need to subtract $\frac{10}{2}$ with and $9.3808/2$ we get a very small digit.

So this results in cancellation and

we obtain loss of significance

$$\rightarrow x_1 = 10 - 5 + \frac{\sqrt{88}}{2} \rightarrow 9.69$$

$$\rightarrow x_2 = 5 - \frac{\sqrt{88}}{2} \rightarrow 0.31 \rightarrow \text{Loss of significance}$$

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(c)

We are required to discuss an alternative approach to avoid loss of significance.

$$x_1 = \frac{10 + \sqrt{88}}{2} \approx 9.6904 \quad [4sf]$$

We know the equation is,

$$ax^2 + bx + c = 0$$

$$(\text{Root 1}) (\text{Root 2}) = \frac{c}{a}$$

$$(x_1) (x_2) = \frac{c}{a} = \frac{3}{1}$$

$$x_2 = \frac{3}{9.6904} \approx 0.3096$$

So, we can avoid loss of significance.